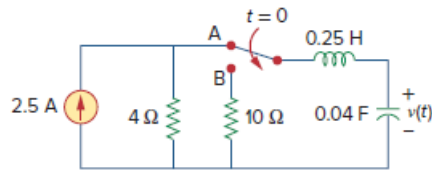


Problem

In the circuit of Fig. 16.44, the switch moves (make before break switch) from position *A* to *B* at $t = 0$. Find $v(t)$ for all $t \geq 0$.

Figure 16.44:



Step-by-step solution

Step 1 of 7

Refer to the circuit diagram of Figure 16.44 in the textbook.

The switch stays at position *A* for a long time.

Thus, the circuit is in steady state at $t = 0^-$.

In steady state inductor is short circuited and capacitor gets open circuited.

The modified circuit diagram in steady state for $t = 0^-$ is shown in Figure 1.

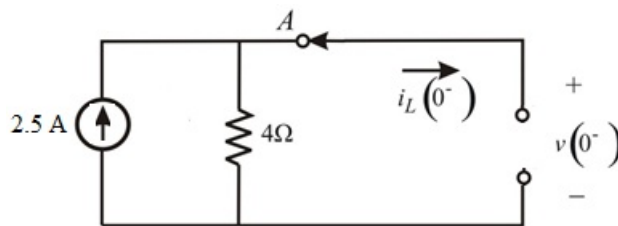


Figure 1

Step 2 of 7

Calculate the initial voltage $v(0^-)$ across the capacitor.

$$\begin{aligned} v(0^-) &= (2.5)(4) \\ &= 10 \text{ V} \end{aligned}$$

From Figure 1, the initial inductor current $i_L(0^-)$ is,

$$i_L(0^-) = 0 \text{ A}.$$

In steady state, $i_L(0^-) = i_L(0^+) = i_L(0)$ and $v(0^-) = v(0^+) = v(0)$. Hence,

$$\begin{aligned} i_L(0) &= 0 \text{ A} \\ v(0) &= 10 \text{ V} \end{aligned}$$

Step 3 of 7

At $t = 0$ switch is moved to position B.

Redraw the circuit in Figure 1 for $t > 0$ with the initial values.

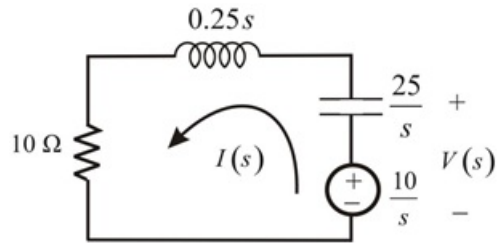


Figure 2

Step 4 of 7

Apply Kirchhoff's voltage law to the loop $I(s)$.

$$-\frac{10}{s} + \frac{25}{s}I(s) + 0.25sI(s) + 10I(s) = 0$$

$$I(s) \left[\frac{25}{s} + 0.25s + 10 \right] = \frac{10}{s}$$

$$I(s) \left[\frac{1}{4}s^2 + 10s + 25 \right] = 10$$

$$I(s) = \frac{40}{s^2 + 40s + 100}$$

Step 5 of 7

Calculate the voltage across the capacitor.

$$V(s) = (0.25s + 10)I(s)$$

Substitute $\frac{40}{s^2 + 40s + 100}$ for $I(s)$ in equation.

$$\begin{aligned} V(s) &= (0.25s + 10) \left[\frac{40}{s^2 + 40s + 100} \right] \\ &= \frac{10s + 400}{s^2 + 40s + 100} \end{aligned}$$

Decompose $V(s)$ into simple terms using partial fraction method.

$$V(s) = \frac{A}{(s + 2.679)} + \frac{B}{s + 37.32} \dots\dots (1)$$

$$\frac{10s + 400}{s^2 + 40s + 100} = \frac{A}{(s + 2.679)} + \frac{B}{(s + 37.32)}$$

Step 6 of 7

Find coefficient A.

$$\begin{aligned} A &= (s + 2.679)V(s) \Big|_{s=-2.679} \\ &= \frac{10s + 400}{(s + 37.32)} \Big|_{s=-2.679} \\ &= \frac{(10)(-2.679) + 400}{(-2.679 + 37.32)} \\ &= 10.77 \end{aligned}$$

Find coefficient B .

$$\begin{aligned} B &= (s + 37.32)V(s) \Big|_{s=-37.32} \\ &= \frac{10s + 400}{(s + 2.679)} \Big|_{s=-37.32} \\ &= \frac{(10)(-37.32) + 400}{(-37.32 + 2.679)} \\ &= -0.774 \end{aligned}$$

Substitute 10.77 for A and -0.774 for B in equation (1).

$$V(s) = \frac{10.77}{(s + 2.679)} - \frac{0.774}{(s + 37.32)}$$

Laplace transform pair:

$$e^{-at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s + a}$$

Apply inverse Laplace transform on both sides of equation $V(s)$.

$$v(t) = 10.77e^{-2.679t}u(t) - 0.774e^{-37.32t}u(t) \text{ V}$$

Therefore, the voltage $v(t)$ across the capacitor is

$$\boxed{\left(10.77e^{-2.679t} - 0.774e^{-37.32t}\right)u(t) \text{ V}}.$$